

Hui (Betty) Zeng

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Education

Harvard/Sloan/Stanford/Wharton Business School, USA **Sept 2028**
Master of Business Administration (MBA)

Technical University of Munich, Munich, Germany **Jan 2024**
M.Sc. Data Engineering and Analytics, (GPA: 1.6 / 1.0, top 5%, German grading scale)

- Erasmus exchange 2022-2023: Katholieke Universiteit Leuven, Belgium – GPA: 17.36 / 20 (Belgian grading scale)
- Master's thesis: *Multimodal 3D Object Detection using Graph Neural Networks*
- Awards: BMW Master Thesis Award, IAPP Westin Scholar

University of Stuttgart, Stuttgart, Germany **Apr 2020**
B.Sc. Geodesy and Geoinformatics, (GPA: 1.8 / 1.0, top 1%, German grading scale)

- Bachelor's thesis: *Human Tracking using Multisensory Data for Mobile Service Robots*

Wuhan University, Wuhan, China **Apr 2020**
B.Eng. Geodesy and Geomatics, (Double Degree, GPA: 3.89 / 4.0, top 3%, Chinese grading scale)

Applied AI & Engineering Experience

Applied AI Consultant **Frankfurt, Germany**
d-fine GmbH *June 2024 – Present*

- **LLM-based agentic Chatbot for Fiber Rollout Planning – Telecom**
 - Developed a QGIS-integrated agentic chatbot with RAG-based document lookup and guided warning resolution, serving 2500+ fiber planners in structured fiber rollout planning across Germany, cutting manual document lookup or rollout computation by +50%.
 - Designed a standardized exception-handling and logging architecture, improved observability and reduced unhandled errors by 30%, raising success rate of served queries to 87% and user reliability to 85%.
- **AI-based Automatic Asset-Object Detection and Extraction from Grid Maps – Telecom**
 - Deployed end-to-end extraction of 6 asset types from ~4.2M grid maps, reducing the manual digitization time per map by 90% and enabling scalable processing across 95 heterogeneous databases in Germany.
 - Delivered asset-specific detection pipelines (e.g. PaddleOCR, YOLO, SegFormer), achieving ~92% precision and ~87% recall on average.
 - Implemented a QGIS Quality Assurance plugin with confidence-based prioritization, integrating ML outputs into existing operations workflows and making data validation >60% more efficient.

Computer Vision Research Intern – Autonomous Driving **Munich, Germany**
BMW Group *May 2023 – Dec 2023*

- Designed a graph-based neural architecture for multimodal 3D perception in autonomous driving, enabling flexible and scalable fusion of diverse sensor configurations. Work patented and recognized with BMW Thesis Award.
- Developed 3 multimodal detector prototypes, dockerized GNN training, outperforming baseline models in moderate & difficult urban scenarios by ~16% in 3D average precision.
- Benchmarked and ran 15+ ablation studies on prototypes using the KITTI dataset, quantifying contribution factor in detection accuracy, computational efficiency, flexibility and scalability.

NLP/GenAI Project Student – Public Sector **Munich, Germany**
TU Munich, School of Computation, Information and Technology *Apr 2023 – Aug 2023*

- Built an end-to-end GenAI application to collect 50k+ German police reports and generate 3k+ synthetic variants, enabling training of a BERT-based detector with 93.16% F1 in distinguishing neural vs real reports.
- Fine-tuned 2 pre-trained LLM models (multilingual T5, GPT2) to diversify text generation, increasing the robustness of the classifier by 6%.
- Analyzed model explainability with shapely values and evaluated machine-generated reports against consistency, fluency, and readability metrics.

Computer Vision Research Intern – Healthcare **Munich, Germany**
LMU Clinic and Polyclinic for Dermatology and Allergology *May 2022 – Nov 2022*

- Designed a multimodal skin cancer detector combining images and patient metadata, improving accuracy by 7.2% and recall by 9.3% over image-only baselines and better approximating clinician diagnostic behaviors.
- Closely engaged with dermatologists and co-identified 40+ key indicators for skin lesion detection, increasing true positive detection in Seborrheic Keratosis (>30%) and Actinic keratoses (>20%).

Selected Professional & Project Experience

Algorithmic Trading Strategies

Online, UK

Certificate for Quantitative Finance

Jun 2025 – June 2026

- Designed and validated a systematic algorithmic trading system on live equity data (Micron Technology), engineering 15+ trend and mean-reversion indicators across clock-, volume- and dollar-time bar representations and screening signal quality via Information Coefficient significance testing.
- Built a rigorous statistical validation pipeline (walk-forward analysis, random-timing significance benchmark, transaction-cost sensitivity sweep) that exposed false-positive strategies and isolated a genuine, tradeable trend-following edge, delivering a diversified dual-timeframe portfolio with Sharpe ratio 2.43 and a 28% reduction in maximum drawdown (−9.9% to −7.1%) versus the single-strategy baseline.
- Deployed the strategy end-to-end via a broker API (Alpaca) with automated live execution, order-fill verification, rolling VaR/drawdown risk monitoring, signal-drift detection and Slack alerting, containerized with Docker for continuous unattended operation.

Education Intern - Digital Education

Paris, France

UNESCO

Mar 2024 – June 2024

- Launched a new website and SEO-strategy for UNESCO's education initiative Global Skills Academy, increasing engaged potential learners by ~35% for free AI/digital, green and entrepreneurial trainings.
- Collaborated with 25 business partners, UNEVOC and TVET centres to conduct skill demand analysis and design market-entry strategies, communicating analytics insights to education and policy stakeholders.
- Built the initiative's first monitoring framework with PowerBI, defining qualitative and quantitative KPIs (engagement, conversion, learning progress, satisfaction).

Privacy Research Student

Leuven, Belgium

KU Leuven, Computer Security and Industrial Cryptography

Sep 2022 - Feb 2023

- Initiated and led a 3-person team for a privacy impact assessment project on Tinder, coordinating scope, research tasks and delivery. Paper recognized with IAPP Westin Scholar Award.
- Assessed the platform's end-to-end data lifecycle and AI algorithms for privacy and compliance, and communicated findings to both technical and non-technical audiences in written and oral form.
- Identified key privacy risks in anonymization, encryption and centralized ML, and presented concrete recommendations to improve privacy while maintaining utility.

Software Engineering Intern

Stuttgart, Germany

Helsinki Systems GmbH

Aug 2021 – Oct 2021

- Co-developed a full-stack, card-based COVID contact-tracking system for 50k+ staff at the University of Stuttgart, spanning SQL data model, backend web APIs and card-reader integration.
- Implemented and deployed Python backend services for data ingestion, querying, analysis and status visualisation, enabling real-time campus health monitoring and contact tracing.

Awards & Scholarships

Westin Scholar Award 2023, International Association for Privacy Professionals

the only nominee by KU Leuven for excellence in privacy, cybersecurity, or related technology fields.

Master Thesis Award 2023, BMW Group

awarded to outstanding Master thesis in academic and applied science research.

Scholarship for International Organization Program 2024, China Scholarship Council

merit-based sponsorship for work at UNESCO

Skills & Certificates

GenAI & LLMs: LLMs, prompt engineering, fine-tuning, RAG, agentic workflows, text classification, explainability

Machine Learning: Supervised learning, deep learning, multimodal models, Computer Vision, 3D perception (LiDAR), GNNs, evaluation

Programming Languages & Frameworks: Python (Pytorch, TensorFlow, scikit-learn, pandas, LangChain, LangGraph), R, SQL, C++, JavaScript, RestAPI, Django

Tools & Infrastructure: Docker, MLflow, Azure, Google Cloud Platform, Linux, Git, QGIS

Languages: Chinese(native), Cantonese(native), English(bilingual), German(C1), French(A1)

Certificates: IAPP Certified Information Privacy Technologist, Azure Data Scientist Associate, Certificate for Quantitative Finance